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Lumbar spine movement patterns during prolonged sitting differentiate low back pain developers from matched asymptomatic controls

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Abstract. *Background:* Little is known about how lumbar spine movement influences mechanical changes and the potential injurious effects of prolonged flexion associated with seated postures. The purpose of this study was to examine the postural responses and pain scores of low back pain sufferers compared with asymptomatic individuals during prolonged sitting in order to understand the biomechanical factors that may be associated with sitting induced low back pain.
Method: Sixteen participants with sitting-aggravated low back pain were age- and gender-matched with 16 asymptomatic participants. Tri-axial accelerometers were used to monitor lumbar spine angles during 90 minutes of seated computer work. Lumbar spine postures were examined using a movement pattern analysis of two types of postural adjustments, termed shifts (step-like adjustments larger than 5°) and fidgets (small change and return to approximately the same position).
Results: The LBP group reported large significant increases ($P < 0.0001$) in low back pain while asymptomatic individuals reported little to no pain. On average, every participant fidgeted every 40 to 50 seconds. However, only the LBP sufferers demonstrated a significant increase ($P = 0.04$) in the number of shifts over 90 minutes of seated work; the LBP group shifted every 4 minutes in the last 30 minutes of sitting compared to every 10 minutes for the asymptomatic group. LBP sufferers also demonstrated larger amplitudes of shifts and fidgets when compared to the asymptomatic group.
Conclusion: Greater and more frequent movement was not beneficial and did not reduce pain in individuals with pre-existing LBP. Future work to understand the biomechanical effects of proactively inducing movement may help to explain the paradox of the relationship between movement and pain.

Keywords: Lumbar spine, pain, seated work, movement

1. Introduction

The relationship between workplace exposure to prolonged sitting and low back pain (LBP) is controversial; systematic reviews of the literature fail to show convincing evidence that LBP is strongly linked to prolonged sitting [25,32]. However, the fact remains that populations who have LBP that is induced and/or aggravated by prolonged sitting do indeed exist [15, 41,56]. Furthermore, subjective experimental findings demonstrate that low back discomfort increases with prolonged seated exposures, even in individuals with no prior history of LBP [7,18,22,23]. A move towards the examination of time-varying responses of the lumbar spine is necessary in order to determine the mechanisms for sitting-induced LBP in situations rel-

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evant to workplace exposures. Little is known about how movement in the lumbar spine influences the mechanical changes and potential injurious effects of the prolonged flexion associated with seated postures and whether movements are preventative of or in response to discomfort/pain. Thus we were driven to document the seated movement patterns of a group of individuals with a history of LBP induced or aggravated by prolonged sitting and compare them to an asymptomatic group.

Several research attempts have increased the understanding of the biomechanical differences between individuals with LBP and asymptomatic individuals and how these factors affect seated postural and muscular responses. Previous research shows that LBP status can affect lumbar spine and pelvic postures in sitting. Individuals with flexion-provoked pain sit closer to lumbar spine flexion end range with greater posterior pelvic tilt [41] and this results in a reduced "usable" lumbar range of movement in sitting [15]. Furthermore, LBP patients with skeletal asymmetries compensate with altered motion in both the thoracic and lumbar regions during seated movements [5]. Several studies demonstrate impaired trunk postural control in individuals with LBP that could be caused by poor proprioception in the lumbar spine [40,42]. In fact, LBP patients appear to lack the ability to produce fine postural adjustments in both flexion/extension and lateral bending and they have difficulty repositioning their lumbar spine into a neutral posture [40,42]. In seated postures, LBP patients demonstrate higher back muscle activation [14] and delayed muscle responses to sudden loading [42]. The increased muscle activity is consistent with a protective reflex at the end range of lumbar spine flexion [47]. It is thought that these altered muscle coordination patterns are aimed at protecting the lumbar spine in order to avoid or reduce pain. However, if this guarding behaviour persists, it may exacerbate the pain or lead to new complaints [33]. Co-contraction will increase forces acting on the spine and hyperactivity in the muscles could lead to spasm and more pain and limit the functional abilities of LBP patients [51].

Most studies have examined a static snapshot of different sitting conditions to observe representative muscle activity, posture, and loading. Seated office work can involve a wide range of postures of up to 50% of lumbar spine range of motion (RoM) with frequent movements [13,50]. Thus, a shift towards the examination of the time-varying responses of the lumbar spine is necessary in order to determine the mechanisms for LBP generation and how certain variables can be manipulated to avoid or prolong the initiation of these mechanisms. Jensen and Bendix [28] suggested that there are three factors contributing to LBP in sitting: 1) insufficient nutrition to the intervertebral disc; 2) the stress-relaxation of spinal ligaments and; 3) muscular fatigue. Therefore, it is necessary to establish a more complete knowledge of how movement of the lumbar spine affects pain generating mechanisms in sitting. Sitting specific movement patterns of the lumbar spine need to be established in order to attempt to understand the relationship between lumbar spine movement and LBP.

One of the difficulties with prolonged examinations is finding ways to quantify time-varying data. Several previous approaches attempted to document postural changes during prolonged sitting including: counting the number of times that whole body posture changed [9,29,31]; seat pan centre of pressure (CoP) movements [21,22]; variation of lumbar spine postures quantified using the amplitude probability distribution function (APDF) [13]; and global postural (macromovements) and small "fidgeting" changes (micromovements) in lumbar spine curvatures using a flexible goniometer [52,53]. In a related technical approach, Duarte and Zatsiorsky [17] developed algorithms to detect different migration patterns of the centre of pressure under the feet during prolonged standing. A similar approach was developed for the current study in order to quantify lumbar spine movement.

In order to understand the relationship between lumbar spine movement and LBP, it is necessary to first examine the sitting behaviour of individuals with sitting-induced LBP. Thus, the purpose of this study was to examine any association between postural responses and pain development in a LBP population and compare it to an asymptomatic control population in order to understand the biomechanical factors associated with LBP during sitting. Specific goals included: (1) examining the postural responses and pain scores of LBP and asymptomatic individuals to 90 minutes of seated computer work; (2) to develop different methods of quantifying lumbar spine movement patterns and to determine if it is possible to distinguish between LBP and asymptomatic groups based on postural responses.

2. Methods

2.1. Participants

A total of thirty-two participants were recruited from the community at large. Eight asymptomatic males and

Table 1
Participant characteristics and Oswestry Disability Index (ODI) scores taken at the beginning of the testing session. Significant differences between the asymptomatic and LBP groups were seen in ODI score only (level of significance indicated by P-value)

	Age (years)	Height (m)	Mass (kg)	BMI (kg/m^2)	ODI Score (%)
Female					
Asymptomatic ($n = 8$)	25.9 (4.6)	1.67 (0.06)	64.4 (13.3)	23.0 (4.7)	1.1 (2.0)
LBP ($n = 8$)	27.0 (5.3)	1.70 (0.06)	70.7 (13.3)	24.6 (5.3)	17.5 (9.2)
	n/s	n/s	n/s	n/s	$P = 0.002$
Male					
Asymptomatic ($n = 8$)	24.0 (3.8)	1.80 (0.10)	74.0 (11.5)	23.3 (3.3)	0.3 (0.8)
LBP ($n = 8$)	22.8 (3.2)	1.78 (0.05)	81.3 (17.3)	24.8 (3.4)	14.2 (5.2)
	n/s	n/s	n/s	n/s	$P = 0.0001$

eight asymptomatic females with no history of LBP for 12 months prior to the testing period were age- and gender-matched to eight males and eight females with sitting-induced LBP (Table 1). Asymptomatic participants were included if they had no self-reported activity-limiting LBP over the 12 months prior to the testing period. Inclusion criteria for LBP participants were: non-specific back pain with insidious onset for longer than 2 months (less than 2 years), self-reported lumbar pain and discomfort elicited by prolonged sitting and minimal pain at the time of testing (i.e. at the start of the test, participants were not having an episode of LBP that would prevent them from sitting for 90 minutes). LBP participants were excluded if they had been previously diagnosed with neurological deficits and/or lower extremity impairments (i.e. numbness, weakness, decreased reflex), stenosis, spondylolisthesis, recent fractures, severe structural deformities (kyphosis or scoliosis) or had previous surgical intervention. All participants (both asymptomatic and LBP) completed the Oswestry Disability Index 2.0 [20] at the beginning of the test session prior to sitting in order to ascertain how LBP affected their ability to manage in everyday life (Table 1). An ODI score of 0 to 20% indicates minimal disability and 21 to 40% indicates moderate disability [20]. The study protocol received approval from the University of Waterloo Office of Research and participants gave informed consent before testing began.

2.2. Instrumentation and data collection

Lumbar spine flexion angles were obtained using tri-axial accelerometers (S2-10G-MF, NexGen Ergonomics, Montreal, Quebec, Canada) placed over the L1 spinous process and the sacrum at the level of S2. Accelerometers have been used as inclinometers to measure the spine in quasi-static and static situations [8,24,57] and have been shown to correlate well with a common fin marker method of measuring spine angles [44]. Only flexion-extension was examined in this study because the link between low back pain and sitting has been attributed to the flexed curvature of the lumbar spine [54]. Accelerometer signals were A/D converted at rates of 128 Hz with a 16-bit A/D system (Optotrak Data Acquisition Unit II, Northern Digital Inc., Waterloo, ON, Canada).

Lumbar spine RoM was assessed before and after the 90-minute sitting period. A quiet upright standing trial was collected for 5 seconds. Then participants were asked to bend over and touch their toes while focusing on maximally flexing their lumbar spine and data were collected for 5 seconds while the position was held. The sitting period involved 90 minutes of seated computer work, performed at a workstation that was adjusted so that the elbows were at 90° with relaxed shoulders when each participant was typing on the keyboard. The office chair (Jendra #3614, Borgo Contract Seating, Etobicoke, Ontario, Canada) was adjusted such that the initial seated position allowed the knees to be at 90° when the feet were in full contact with the floor. The seat pan to back rest angle was fixed at an angle of 105° and the seat pan was locked in a position parallel to the floor. Participants were instructed to sit how they would normally and move around in the chair at will with the constraint imposed of no leg crossing. During the 90-minute sitting period, participants performed 15-minute intervals of controlled simulated office work consisting of a mousing task, a typing task and task involving a combination of the two. The three tasks were presented in a random order and then repeated in the same order to ensure that a full data set was obtained should a participant not be able to complete the 90-minute sitting period. Accelerometer data were continuously collected throughout each 15-minute interval.

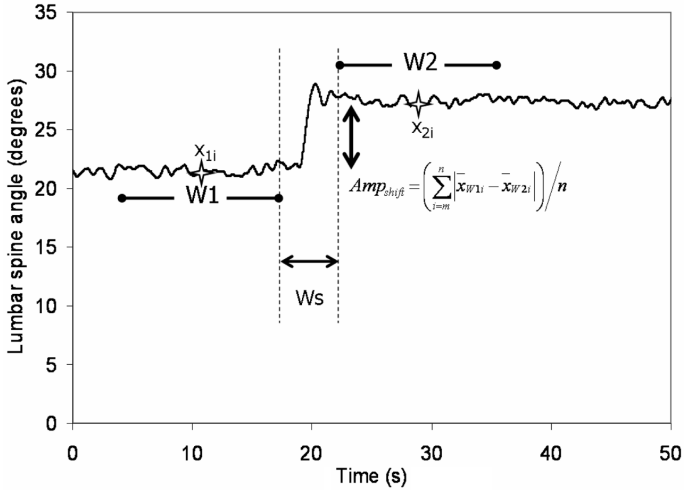


Fig. 1. An example of the shift pattern and the parameters used in the recognition algorithm. The average values were computed on a point-by-point basis for two 15-second moving windows (W_1 and W_2), each centered on a point (x_{1i} and x_{2i}) and separated by W_s (3 seconds). Any two consecutive windows being offset by a change greater than five degrees were classified as a shift. Amp_{shift} indicates the calculation for the shift amplitude.

Participants rated their perceived level of pain on a 10 cm visual analogue scale (VAS) at the start and after each of six 15-minute intervals, yielding a total of seven VAS scores per participant during the 90-minute sitting period. VAS scores were obtained for three regions of the low back: the central area over the lumbar spine, and the right and left sides lateral to the spine. Visual analog scales are generally accepted by the International Association for the Study of Pain as a valid scale for pain and discomfort assessments [10]. Participants were asked to place a mark along the 10 cm line where they felt their current level pain was on a scale of "no discomfort" to "worst discomfort imaginable".

2.3. Data reduction and processing

Given that the outputs from the accelerometers are a sinusoidal function of the inclination of the sensor relative to gravity, each axis was recorded at 0° (or $+1$ g) and 180° (or -1 g) in order to provide a conversion from voltage outputs to degrees. Each accelerometer was oriented so that the positive y-axis was aligned pointing down the vertical axis of the spine. The y-axis was used as the inclination axis and the corresponding orthogonal z-axis was used in order to determine the correct orientation of the y-axis inclination. Because of the sinusoidal nature of the accelerometer output, two axes are necessary in order to determine the correct orientation of the sensor. For example, when the z-axis was oriented to the left (90°), the y-axis was pointed up (180°). The lumbar spine angle was calculated as the difference between the y-axis inclinations of the L1 and sacral sensors. Converted angles were smoothed using a zero-lag 4th order low pass Butterworth filter with a cut-off frequency of 1 Hz to remove any higher frequency accelerations. Lumbar spine RoM was calculated as the change in angle from upright standing to the fully flexed posture in the toe touch RoM test. The mean lumbar spine angles from the pre- and post-sitting RoM trials were compared to determine if there were any effects of prolonged sitting on lumbar spine RoM. The pre-sitting lumbar spine RoM value was used to normalize all lumbar spine angles throughout the 90-minute sitting period. Time-varying lumbar spine angles were quantified using the following methods:

1. The amplitude probability distribution function (APDF) was used to examine the range and time distribution of lumbar spine postures adopted during the 90-minute sitting period [13]. Briefly, the normalized time-varying lumbar spine angles were binned in 1% increments to generate an APDF for each individual.
2. Shift and fidget patterns were determined using the following movement pattern algorithms:

When lumbar spine postures were displayed as a time-series, two patterns of movement were discernible (Figs 1 and 2):

1. Shifts: step-like angular change in lumbar angle
2. Fidgets: small rapid changes of the lumbar angle about the same average position

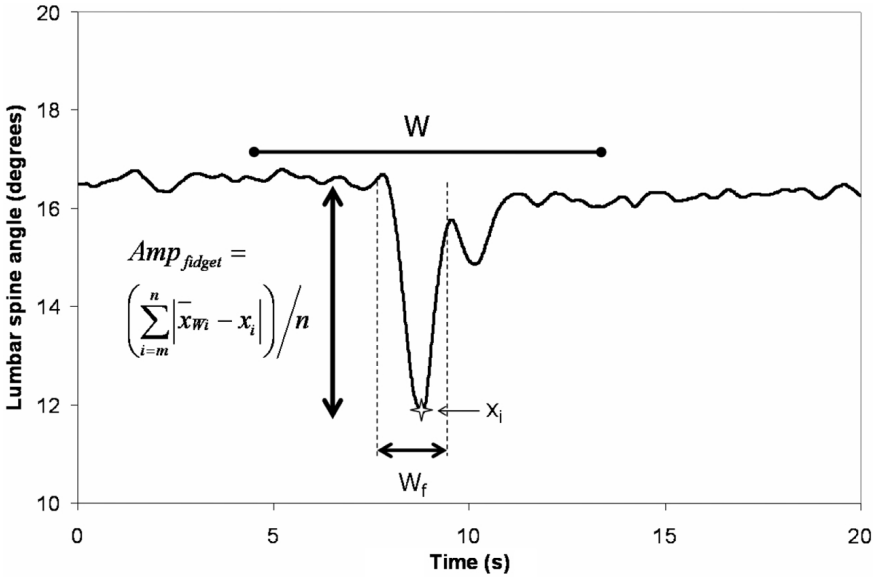


Fig. 2. An example of the fidget pattern and the parameters used in the recognition algorithm. The average and standard deviation were computed on a point-by-point basis for a moving window (W) of length W (60 seconds) and centered on a point (x_i). Amp_{fidget} indicates the calculation for the fidget amplitude.

Two separate algorithms were developed in order to detect these two movement patterns in time-varying lumbar angle.

2.3.1. Shift algorithm

To detect shifts in the lumbar spine angle time series (Fig. 1):

1. The average values ($\overline{x}_{W1i}$ and $\overline{x}_{W2i}$) for two consecutive moving windows, W_1 and W_2 each of 15 seconds in length, centered around a point (x_{1i} and x_{2i}) and separated by the 3 second period W_s, were computed. $\overline{x}_{W1i}$ and $\overline{x}_{W2i}$ were calculated and compared on a point by point basis (where i corresponds to the centre point of the moving window).
2. Any two consecutive windows being offset by a change greater than five degrees (Eq. (1)) were classified as a shift:

$$|\bar{x}_{W1i} - \bar{x}_{W2i}| \geqslant 5^{\circ} \quad (1)$$

3. The shift duration was determined as the length of time Eq. (1) was consecutively satisfied, as long as the shift duration was greater than W_s.
4. The shift amplitude in degrees was calculated as:

$$Amp_{shift} = \left(\sum_{i=m}^{n} |\overline{x}_{W1i} - \overline{x}_{W2i}|\right)/n \quad (2)$$

Where:

$n =$ the total number of consecutive points satisfying Eq. (1) for a given shift
$i =$ frame number where Eq. (1) is satisfied
$m =$ the last consecutive frame satisfying Eq. (1) for a given shift

2.3.2. Fidget algorithm

To detect a fidget in the lumbar spine angle time series (Fig.2):

1. The average and standard deviation values ($\overline{x}_{Wi}$ and SD_{Wi}) of a moving window (W), centered on a point (x_i) were computed. The length of W was 60 seconds.
2. The fidget amplitude threshold value was set as 3*SD_{Wi} and the fidget duration, W_F, was selected as less than 3 seconds in order to distinguish it from a shift Eq. (3).

$$|\overline{x}_W - x_i| > 3SD_W \quad (3)$$

3. The fidget duration, W_F was determined as the length of time Eq. (3) was consecutively satisfied, as long as the time did not exceed 3 seconds.
4. The fidget amplitude was calculated as:

$$Amp_{fidget} = \left(\sum_{i=m}^{n} |\overline{x}_{Wi} - x_i|\right)/n \quad (4)$$

Where:

$n =$ the total number of consecutive points satisfying Eq. (3) for a given fidget
$i =$ frame number where Eq. (3) is satisfied
$m =$ the last consecutive frame satisfying Eq. (3) for a given fidget

2.4. Statistical analyses

Pre- and post-sitting lumbar spine angles, the time-varying changes in VAS scores, and number of shifts and fidgets were compared using a mixed model 2-way analysis of variance (ANOVA) (LBP status, time) with repeated measure on one factor (time). Baseline VAS scores were compared using a 2-way ANOVA (LBP status, body region). Average shift and fidget amplitudes and frequencies were compared between LBP and asymptomatic groups using Student's t-tests. Tukey's post-hoc multiple comparisons were used to examine significant main effects and interactions. The level of significance was set at $P < 0.05$.

3. Results

3.1. VAS scores

Baseline VAS scores for the three low back regions were not significantly different and were pooled to give an overall low back pain score when comparing the LBP and asymptomatic groups. The difference between baseline VAS scores was significant between the groups ($P = 0.01$); the asymptomatic group's baseline score was 0.1 mm (SD 0.1) and the LBP baseline score was 9.6 mm (SD 3.8). There was a significant LBP status*time interaction ($P < 0.0001$) for low back VAS scores in all three regions; scores increased significantly over time for the LBP group, while remaining relatively close to zero for the asymptomatic group (Fig. 3).

3.2. Lumbar spine range of motion

Lumbar spine RoM was significantly different ($P = 0.04$) between the LBP group that had an average RoM of 47.4° (SD 14.0) and the asymptomatic group with an average of 58.0° (SD 15.1). There was no significant difference between the groups in pre and post sitting lumbar spine RoM; however in both groups, participants lost an average of 2.4° (SD 4.1) of lumbar spine RoM after 90 minutes of sitting ($P = 0.003$).

3.3. Time-varying lumbar spine movement patterns

There were two patterns of seated lumbar spine postures adopted by the asymptomatic and the LBP groups, as demonstrated by the APDF (Fig. 4). Asymptomatic individuals generally adopted a more "static" sitting strategy where they used a smaller range of their lumbar spine RoM over the 90-minute sitting period. LBP individuals generally adopted a wide range of postures over a larger RoM (Fig. 4). The range of movement of the lumbar spine was determined using the maximum and minimum lumbar spine angles achieved during the sitting period. Asymptomatic individuals used 26.5% (SD 9.8) of their RoM on average while LBP individuals used an average of 61.4% (SD 27.3), which were significantly different ($P < 0.0001$).

Over the entire sitting period, LBP sufferers shifted once every 7.2 ± 4.7 minutes, whereas healthy participants shifted every 12.6 ± 4.6 minutes and the difference in shift frequency was statistically significant ($P = 0.003$; Table 2). Every participant demonstrated some level of a fidget movement pattern, where a fidget occurred on average every 40 to 50 seconds (Table 2). However, the amplitudes of both the shift and fidget movement patterns of the LBP group were 1.5 and 2.5 times larger than the asymptomatic group (Table 2). There was also a significant LBP*Time interaction ($P = 0.04$) whereby the LBP group shifted more times than the asymptomatic group in the last three 15-minute time blocks (Fig. 5).

4. Discussion

Time-varying lumbar spine movement patterns distinguished between the LBP and asymptomatic groups during prolonged seated office work. The major finding of this study was that LBP sufferers moved more than asymptomatic individuals during 90 minutes of seated work and they reported increased LBP over time. LBP sufferers used as much as 80% of their lumbar spine RoM while sitting and demonstrated large, frequent shifts in lumbar spine posture. These movement patterns increased in amplitude and frequency over time along with LBP development. On the other hand, asymptomatic individuals generally exhibited a more static sitting pattern, using only 30% of their lumbar RoM with smaller fidget movements around a mean posture and reported little to no LBP development. For asymptomatic individuals, seated movements may serve a purpose to diminish feelings of dis-

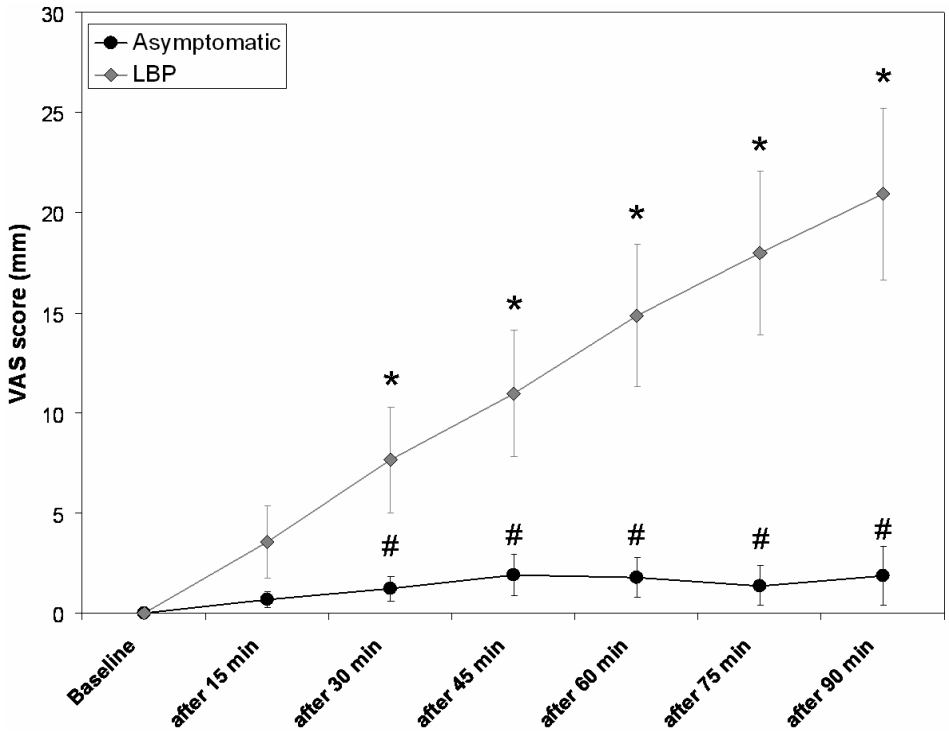


Fig. 3. Average ($\pm$ 1 SD) Visual Analogue Scale (VAS) scores for the low back over 90 minutes of sitting for the low back pain ($n = 16$) and asymptomatic (n =16) groups. There was a significant LBP*time interaction ($P < 0.0001$). The asterisk (*) indicates that the LBP VAS score is significantly different from baseline. The number sign (#) indicates that the LBP VAS score is significantly different from the asymptomatic group.

comfort from restlessness [9,29], body stiffness, lack of circulation [55] or seat pressure [16]. In the current study, larger magnitude and more frequent movement was not beneficial and did not reduce or prevent pain in individuals with pre-existing LBP.

Both LBP and asymptomatic individuals in the current study demonstrated a decrease in lumbar spine RoM after 90 minutes of sitting, which is likely synonymous with an increase in lumbar spine stiffness. Previous research has suggested that increased spine stiffness after prolonged sitting could occur because spinal movement may increase spinal height through fluid flow into the disc [7], especially if a chair back rest is present [50]. However, the increased stiffness in the LBP and asymptomatic individuals in the current study may be due to different mechanisms. Due to the less dynamic nature of LBP-free sitting, increased spinal stiffness could be due to changes in passive elastic properties of the muscles and passive tissues surrounding the joint. However, chronic LBP patients generally exhibit higher back muscle activation, most likely aimed at avoiding painful tensile stresses placed on spinal structures [51]. Even slight increases in trunk extensor activity associated with small postural changes [6,12] or low back pain disorders [51], could have negative effects on the muscles of the back, thus increasing lumbar spine stiffness. This study lacks information regarding muscular activity during prolonged sitting, which may have helped explain the impact of the movement patterns and subsequent post-exposure response seen in the different groups.

Other studies have quantified trunk and lumbar spine movements of asymptomatic populations using various methods. Two sitting strategies in an asymptomatic group were previously described based on the amplitude probability distribution function (APDF) of time-varying lumbar spine postures [13]. "Static" sitters used less than 10% of their total lumbar flexion RoM whereas "dynamic" sitters used up to 50% of their RoM, but no relationship between sitting strategies and discomfort was established. The results from our study demonstrated that asymptomatic individuals generally adopted the "static" sitting strategy, while LBP individuals were "dynamic" sitters. Vergara and Page [52, 53] noted that increases in lumbar discomfort reporting were associated with less frequent lumbar fidgeting and

Table 2
Lumbar spine shift and fidget analysis. Shifts are defined as a step-like change in lumbar spine angle. Fidgets are defined as a change and return of lumbar spine angle to approximately the same position. Significant differences between the low back pain (LBP) and asymptomatic groups are indicated by asterisks (*)

	Healthy	LBP	*P*-value
Shift amplitude (normalized to %RoM)	11.2% (5.2)*	17.3% (6.7)*	$P = 0.01$
Shift frequency (# minutes between shifts)	12.6 min (4.6)*	7.2 min (4.7)*	$P = 0.003$
Fidget amplitude (normalized to %RoM)	4.4% (3.6)*	10.9% (8.0)*	$P = 0.001$
Fidget frequency (# seconds between fidgets)	47.0 s (9.2)	42.1 s (9.9)	n/s

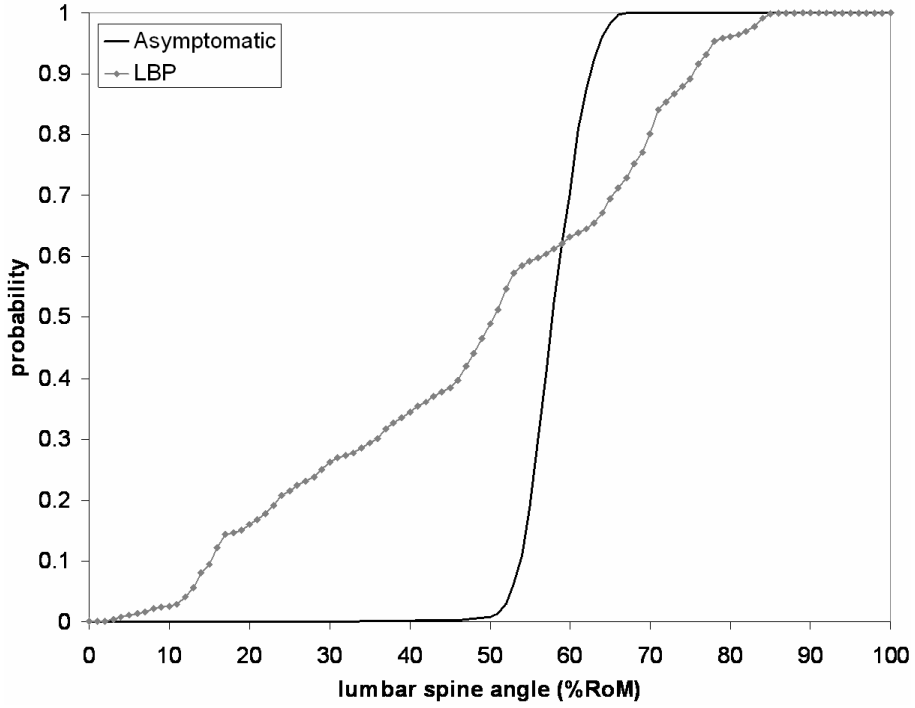


Fig. 4. Amplitude probability distribution functions of lumbar spine angles for one asymptomatic and one LBP participant for the duration of the sitting period.

more frequent lumbar and pelvic shifts. They suggested that small movements around a mean posture help to reduce muscular strain while large changes in posture are a good indicator of the presence of discomfort. Based on their quantification methods, they documented large lumbar spine shifts (or "macromovements") that ranged from one change every four to eight minutes with amplitudes from 14 to 27°. "Micro-movements" occurred every 4.5 to 6.5 seconds, with amplitudes of 1 to 2.5°. In the present study, the definition of a shift was very similar to that of Vergara and Page [53] but the shift amplitudes never exceeded 13° (or 40% RoM). This is likely due to the mathematical calculations of this study's method, which compared average values of two moving windows, thus perhaps masking any large peaks in lumbar spine posture. Therefore, large peaks in lumbar spine movement could be occurring that may be more detrimental to spine health than beneficial as higher trunk velocity has been identified as a risk factor for LBP [34]. Future work could be aimed at examining the efficacy of slower, more controlled movements in the relief of sitting-induced LBP.

One of the uncertainties of the algorithm used to identify shifts in the current study is the choice of the shift amplitude threshold of 5°. This value was chosen because it has been used as a threshold by other re-

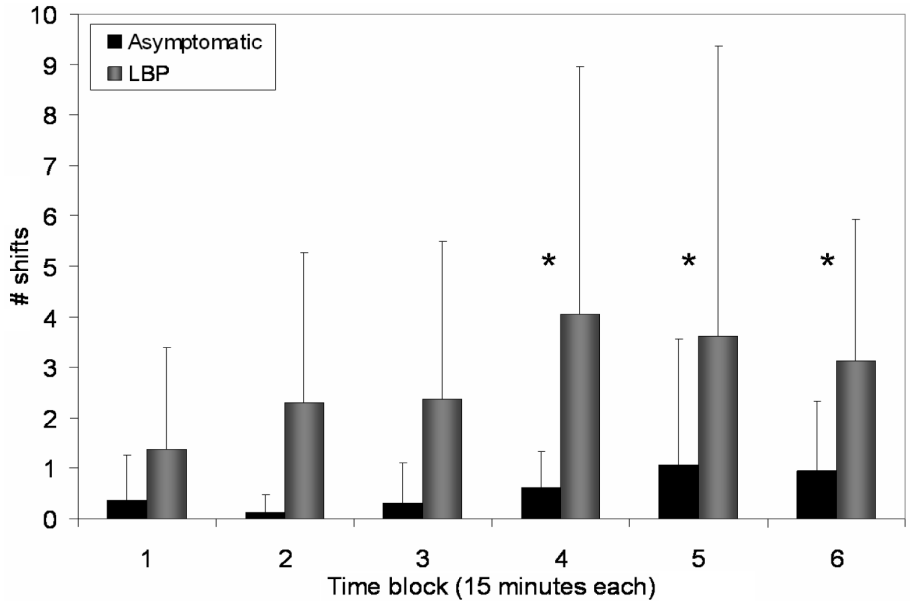


Fig. 5. The average (+ 1SD) number of shifts during 90 minutes of sitting for each 15-minute time block for the two test groups (low back pain and asymptomatic). There was a significant LBP*time interaction (P = 0.04) whereby LBP group had significantly more shifts in the last 3 time blocks than the asymptomatic group. The asterisk (*) indicates a significant difference between LBP and asymptomatic groups ($P < 0.05$).

searchers to define large lumbar spine movements [52, 53] and because flexion at the L4/5 joint contributes approximately 40% to the total flexion angle in sitting [19]. Based on this percentage, a 5° change in total lumbar spine angle will effect a 2° change at the L4/5 intervertebral joint (IVJ). While it is not fully understood how much of an angular change will affect the IVJ, 2° increment changes of a spinal segment have been shown to affect the mean intervertebral disc pressure [39]. The algorithm employed in the current study to detect a fidget was very different from previous studies [52,53] where the signal was conditioned (i.e. high pass filtered) to isolate the high frequencies. Zero-crossings of the filtered signal were counted as "micromovements". Our method of detecting fidgets used the unaltered time-varying angle signal and the variability of the signal as a threshold to detect biologically meaningful changes in lumbar spine posture. Given the choice of the parameters to mathematically identify a fidget (i.e. fidget threshold, moving window length) there may have been higher frequency movements that were not detected. However, Duarte and Zatsiorsky [17] identified high frequency movement patterns in standing CoP, which they termed trembling. Trembling may have more pertinent implications for studying postural control mechanisms, which was not the focus of our study.

Previous research has suggested that LBP sufferers assume sitting postures with more lumbar spine flexion and posterior pelvic tilt [41]. Contrary to these findings, which were developed based on static analyses of instantaneous seated postures, the development of LBP in the current study was not associated with any one posture. One implication of this finding is that short duration investigations of seated postures may not accurately represent the time varying biological responses to prolonged exposure. Every participant in the current study demonstrated a fidgeting movement pattern, where a fidget occurred on average every 40 to 50 seconds. However, only the LBP sufferers demonstrated a significant increase in the number of shifts over 90 minutes of seated work. In fact, 80% of the asymptomatic individuals did not shift their lumbar spine until after 45 minutes of sitting and fidgeting was the primary movement pattern observed. On the other hand, only 2 LBP sufferers did not shift during the first 45 minutes. These findings support the notion that more frequent shifts are associated with LBP development. It is difficult to address the issue of why individuals moved more over 90 minutes of sitting when so little is understood about why seated individuals move at all. The relationship between in-chair movement and discomfort is elusive since movement is necessary to avoid detrimental static postures. From a tissue perspective, prolonged static trunk postures can have severe detrimental effects on the various structures of the spine. Static flexion can lead to changes in the intervertebral disc (IVD), such

as increased stresses in the annulus of the IVD [4,27] and a loss of disc height over time [11,26]. Furthermore, prolonged bouts of extreme static flexion have been shown to increase joint laxity due to creep of the posterior passive structures of the spine [35] and this creep has the potential to alter spine muscle activity patterns [46]. Prolonged seated work has also been shown to alter the passive stiffness of the in vivo lumbar spine [7]. Movement during sitting may be necessary to promote disc nutrition through fluid flow [1,2,38], relieve potentially harmful stress concentrations in the disc [3,4,39], vary muscle activity [13,36,37] and help circulate blood to and from the muscles [45].

It is difficult to prove a cause and effect relationship between seated spine movement and LBP because the etiology of sitting-induced LBP is generally unknown. There are two plausible explanations as to why movement may not reduce pain in LBP individuals. First, LBP individuals may possess some underlying pathology that is aggravated by increased movement in sitting. LBP sufferers may demonstrate asymmetric movements [5], they may be at their end range of motion already [41], or have altered muscle activation and load sharing [49]. Movement in sitting may lead to strain on tissues that are already vulnerable or reduce spinal stability [51]. Secondly, LBP individuals may move preemptively in order to reduce or prevent expected LBP development. Proactively inducing movement has shown to effectively reduce sitting-induced LBP [48], yet the findings of the current study would argue against this as increased movement did not alleviate LBP development. It is possible that the large and frequent changes in posture observed in LBP sufferers were strategies for preventing pain from becoming worse. Little is known about the biomechanical effects of proactively induced movements. Education regarding movement while sitting, as well as regularly scheduled standing stretch breaks, appears to minimize the increase of low back discomfort over time [21,37]. Furthermore, regular breaks may reduce "static" sitting in asymptomatic individuals and promote muscular load sharing [21,37]. At-work exercise programs are relatively effective at relieving neck/shoulder pain [30], and whole body discomfort [21]. More work is needed to determine whether specific exercises targeting the low back would be appropriate and effective at alleviating sitting-induced LBP. Low frequency, small amplitude continuous passive lumbar motion has also been effective in reducing pain in LBP sufferers over a period of prolonged sitting [43,48]. However, the biomechanical effectiveness of mechanically induced movements, and how these movements can mitigate the detrimental effects of sitting, is largely unknown. Further work to understand the biomechanical effects of proactive induced movement may help to explain the paradox of the relationship between movement and discomfort or pain.

4.1. Conclusions

The proposed methods for identifying shift and fidget movement patterns developed in this work were successful at differentiating between LBP and asymptomatic groups. The results highlight the fact that short duration investigations of seated postures do not accurately represent the time varying biological responses to prolonged exposure. The results from this study also support the theory that large changes in posture are a good indicator of the presence of discomfort in asymptomatic individuals [53] and pain in LBP sufferers. In the current investigation, movement was not beneficial and did not reduce pain in individuals with pre-existing LBP. When one feels movement or a break is necessary, the level of discomfort or pain could be indicating that the process of irritation or damage to the tissues may have already begun. Thus, it is possible that proactively inducing movement may be more beneficial than self-selected movement. However, increased seated movement is not a mechanism that shows potential to reduce LBP development in individuals with sitting-induced LBP. This highlights the need for the development of other interventions in the workplace to help reduce LBP during seated work.

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References

[1] M.A. Adams and W.C. Hutton, The effect of posture on the fluid content of lumbar intervertebral discs, *Spine* 8 (1983), 665-71.

[2] M.A. Adams and W.C. Hutton, The effect of posture on diffusion into lumbar intervertebral discs, *J Anat* 147 (1986), 121-34.
[3] M.A. Adams, D.W. McMillan, T.P. Green, and P. Dolan, Sustained loading generates stress concentrations in lumbar intervertebral discs, *Spine* 21 (1996), 434-8.
[4] M.A. Adams, D.S. McNally, H. Chinn, and P. Dolan, Posture and the compressive strength of the lumbar spine, *Clinical Biomechanics* 9 (1994), 5-14.
[5] E. Al Eisa, D. Egan, K. Deluzio, and R. Wassersug, Effects of pelvic asymmetry and low back pain on trunk kinematics during sitting: a comparison with standing, *Spine* 31 (2006), E135-E143.
[6] B.J. Andersson, B. Jonsson, and R. Ortengren, Myoelectric activity in individual lumbar erector spinae muscles in sitting. A study with surface and wire electrodes, *Scand J Rehabil Med Suppl* 3 (1974), 91-108.
[7] T.A. Beach, R.J. Parkinson, J.P. Stothart, and J.P. Callaghan, Effects of prolonged sitting on the passive flexion stiffness of the in vivo lumbar spine, *Spine J* 5 (2005), 145-54.
[8] E. Bernmark and C. Wiktorin, A triaxial accelerometer for measuring arm movements, *Appl Ergon* 33 (2002), 541-7.
[9] V. Bhatnager, C.G. Drury, and S.G. Schiro, Posture, postural discomfort, and performance, *Hum Factors* 27 (1985), 189-99.
[10] G. Borg, *Borg's Perceived Exertion and Pain Scales*, Champaign, IL: Human Kinetics, 1998.
[11] D.J. Botsford, S.I. Esses, and D.J. Ogilvie-Harris, In vivo diurnal variation in intervertebral disc volume and morphology, *Spine* 19 (1994), 935-40.
[12] J.P. Callaghan and N.M. Dunk, Examination of the flexion relaxation phenomenon in erector spinae muscles during short duration slumped sitting, *Clinical Biomechanics* 17 (2002), 353-60.
[13] J.P. Callaghan and S.M. McGill, Low back joint loading and kinematics during standing and unsupported sitting, *Ergonomics* 44 (2001), 280-94.
[14] W. Dankaerts, P. O'Sullivan, A. Burnett, and L. Straker, Altered patterns of superficial trunk muscle activation during sitting in nonspecific chronic low back pain patients: importance of subclassification, *Spine* 31 (2006), 2017-23.
[15] W. Dankaerts, P. O'Sullivan, A. Burnett, and L. Straker, Differences in sitting postures are associated with nonspecific chronic low back pain disorders when patients are subclassified, *Spine* 31 (2006), 698-704.
[16] M.P. de Looze, L.F. Kuijt-Evers, and J. van Dieen, Sitting comfort and discomfort and the relationships with objective measures, *Ergonomics* 46 (2003), 985-97.
[17] M. Duarte and V.M. Zatsiorsky, Patterns of center of presure migration during prolonged unconstrained standing, *Motor Control* 3 (1999), 12-27.
[18] N.M. Dunk and J.P. Callaghan, Gender-based differences in postural responses to seated exposures, *Clinical Biomechanics* 20 (2005), 1101-10.
[19] N.M. Dunk, A.E. Kedgley, T.R. Jenkyn, and J.P. Callaghan, Seated lumbar intervertebral angles using fluoroscopy, in: *Proceedings of the 31st Annual Meeting of the American Society of Biomechanics*, Stanford, California, August 23-25, 2007.
[20] J.C. Fairbank and P.B. Pynsent, The Oswestry Disability Index, *Spine* 25 (2000), 2940-52.
[21] A. Fenety and J.M. Walker, Short-term effects of workstation exercises on musculoskeletal discomfort and postural changes in seated video display unit workers, *Phys Ther* 82 (2002), 578-89.
[22] P.A. Fenety, C. Putnam, and J.M. Walker, In-chair movement: validity, reliability and implications for measuring sitting discomfort, *Appl Ergon* 31 (2000), 383-93.
[23] D.E. Gregory, N.M. Dunk, and J.P. Callaghan, Stability ball versus office chair: Comparison of muscle activation and lumbar spine posture during prolonged sitting, *Human Factors* 48 (2006), 142-53.
[24] G.A. Hansson, P. Asterland, N.G. Holmer, and S. Skerfving, Validity and reliability of triaxial accelerometers for inclinometry in posture analysis, *Med Biol Eng Comput* 39 (2001), 405-13.
[25] J. Hartvigsen, C. Leboeuf-Yde, S. Lings, and E.H. Corder, Is sitting-while-at-work associated with low back pain? A systematic, critical literature review, *Scand J Public Health* 28 (2000), 230-9.
[26] T.P. Hedman and G.R. Fernie, In vivo measurement of lumbar spinal creep in two seated postures using magnetic resonance imaging, *Spine* 20 (1995), 178-83.
[27] T.P. Hedman and G.R. Fernie, Mechanical response of the lumbar spine to seated postural loads, *Spine* 22 (1997), 734-43.
[28] C.V. Jensen and T. Bendix, Spontaneous movements with various seated-workplace adjustments, *Clinical Biomechanics* 7 (1992), 87-90.
[29] H.W. Jurgens, Body movement during observation tasks, *Ann Physiol Anthropol* 8 (1989), 33-5.
[30] D.M. Kietrys, J.S. Galper, and V. Verno, Effects of at-work exercises on computer operators, *Work* 28 (2007), 67-75.
[31] M.H. Liao and C.G. Drury, Posture, discomfort and performance in a VDT task, *Ergonomics* 43 (2000), 345-59.
[32] A.M. Lis, K.M. Black, H. Korn, and M. Nordin, Association between sitting and occupational LBP, *Eur Spine J* 16 (2007), 283-98.
[33] C.J. Main and P.J. Watson, Psychological aspects of pain, *Man Ther* 4 (1999), 203-15.
[34] W.S. Marras, S.A. Lavender, S.E. Leurgans, F.A. Fathallah, S.A. Ferguson, W.G. Allread, and S.L. Rajulu, Biomechanical risk factors for occupationally related low back disorders, *Ergonomics* 38 (1995), 377-410.
[35] S.M. McGill and S. Brown, Creep response of the lumbar spine to prolonged full flexion, *Clinical Biomechanics* 7 (1992), 43-6.
[36] L. Mclean, M. Tingley, R.N. Scott, and J. Rickards, Myoelectric signal measurement during prolonged computer terminal work, *J Electromyogr Kinesiol* 10 (2000), 33-45.
[37] L. Mclean, M. Tingley, R.N. Scott, and J. Rickards, Computer terminal work and the benefit of microbreaks, *Appl Ergon* 32 (2001), 225-37.
[38] D.W. McMillan, G. Garbutt, and M.A. Adams, Effect of sustained loading on the water content of intervertebral discs: implications for disc metabolism, *Ann Rheum Dis* 55 (1996), 880-7.
[39] D.S. McNally and M.A. Adams, Internal intervertebral disc mechanics as revealed by stress profilometry, *Spine* 17 (1992), 66-73.
[40] P.B. O'Sullivan, A. Burnett, A.N. Floyd, K. Gadsdon, J. Logiudice, D. Miller, and H. Quirke, Lumbar repositioning deficit in a specific low back pain population, *Spine* 28 (2003), 1074-9.
[41] P.B. O'Sullivan, T. Mitchell, P. Bulich, R. Waller, and J. Holte, The relationship beween posture and back muscle endurance in industrial workers with flexion-related low back pain, *Man Ther* 11 (2006), 264-71.
[42] A. Radebold, J. Cholewicki, G.K. Polzhofer, and H.S. Greene, Impaired postural control of the lumbar spine is associated

with delayed muscle response times in patients with chronic idiopathic low back pain, *Spine* 26 (2001), 724-30.
[43] S.M. Reinecke, R.G. Hazard, and K. Coleman, Continuous passive motion in seating: a new strategy against low back pain, *Journal of Spinal Disorders* 7 (1994), 29-35.
[44] J.C. Singer, N.M. Dunk, and J.P. Callaghan, Evaluation of surface marker systems to determine pelvic and lumbar kinematics during sitting, in: *Proceedings of the 4th Annual Ontario Biomechanics Conference*, Barrie, ON, March 9-11, 2007.
[45] G. Sjogaard, G. Savard, and C. Juel, Muscle blood flow during isometric activity and its relation to muscle fatigue, *Eur J Appl Physiol Occup Physiol* 57 (1988), 327-35.
[46] M. Solomonow, R.V. Baratta, A. Banks, C. Freudenberger, and B.H. Zhou, Flexion-relaxation response to static lumbar flexion in males and females, *Clin Biomech* (*Bristol, Avon*) 18 (2003), 273-9.
[47] M. Solomonow, B.H. Zhou, R.V. Baratta, and E. Burger, Biomechanics and electromyography of a cumulative lumbar disorder: response to static flexion, *Clin Biomech* (*Bristol, Avon*) 18 (2003), 890-8.
[48] L.L. van Deursen, J. Patijn, J.R. Durinck, R. Brouwer, J.R. Erven-Sommers, and B.J. Vortman, Sitting and low back pain: the positive effect of rotary dynamic stimuli during prolonged sitting, *European Spine Journal* 8 (1999), 187-93.
[49] J.H. Van Dieen, J. Cholewicki, and A. Radebold, Trunk muscle recruitment patterns in patients with low back pain enhance the stability of the lumbar spine, *Spine* 28 (2003), 834-41.
[50] J.H. Van Dieen, M.P. de Looze, and V. Hermans, Effects of dynamic office chairs on trunk kinematics, trunk extensor EMG and spinal shrinkage, *Ergonomics* 44 (2001), 739-50.
[51] J.H. Van Dieen, L.P. Selen, and J. Cholewicki, Trunk muscle activation in low-back pain patients, an analysis of the literature, *J Electromyogr Kinesiol* 13 (2003), 333-51.
[52] M. Vergara and A. Page, System to measure the use of the backrest in sitting-posture office tasks, *Appl Ergon* 31 (2000), 247-54.
[53] M. Vergara and A. Page, Relationship between comfort and back posture and mobility in sitting- posture, *Appl Ergon* 33 (2002), 1-8.
[54] D.G. Wilder and M.H. Pope, Epidemiological and aetiological aspects of low back pain in vibration environments – an update, *Clinical Biomechanics* 11 (1996), 61-73.
[55] J. Winkel and K. Jorgensen, Evaluation of foot swelling and lower-limb temperatures in relation to leg activity during long-term seated office work, *Ergonomics* 29 (1986), 313-28.
[56] L. Womersley and S. May, Sitting posture of subjects with postural backache, *J Manipulative Physiol Ther* 29 (2006), 213-8.
[57] W.Y. Wong and M.S. Wong, Detecting spinal posture change in sitting positions with tri-axial accelerometers, *Gait and Posture* 27 (2008), 168-71.